

Supplementary Information

Crop-ranking reversal under downside risk and operational constraints

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1 Definitions and complete model notation

Pairwise reversal is $s_i > s_j$, $x_i < x_j$. Complete rank reversal means that the top-ranked crop receives less than every lower-ranked crop. Strong exclusion reversal requires $s_i > s_j$, $x_i = 0 < x_j$. These definitions separate an allocation-order reversal from exclusion of the higher-ranked crop.

Symbol	Definition
$i = 1, \dots, n$	crop index
ω	joint price–yield–cost state

Symbol	Definition
s_i	pre-decision historical relative-yield performance index
x_i	planted area or normalized land share
$\tilde{P}_i, \tilde{Y}_i, \tilde{C}_i$	stochastic price, yield and cost
$\tilde{\pi}_i$	stochastic per-acre margin $\tilde{P}_i\tilde{Y}_i - \tilde{C}_i$
μ_i	expected per-acre margin
$L(x)$	portfolio loss $-\tilde{\pi}^\top x$
A	available land
B, c	working-capital limit and crop cash coefficients
R, r	rotation rows and limits
K, k	contractual-minimum rows and requirements
G, h	shared-capacity rows and limits
ℓ, u	crop-specific lower and upper area bounds
α	loss-CVaR confidence level
κ	loss-CVaR ceiling
ρ	normalized risk-tolerance path index
η	loss-CVaR-ceiling multiplier
ν^A, ν^B, ν^R	land, budget and rotation multipliers
ν^K, ν^G	sign-converted contract and shared-capacity multipliers
ν^ℓ, ν^u	lower- and upper-bound multipliers
$\mathcal{X}$	operational feasible set
$\mathcal{S}_\kappa$	complete optimal set under ceiling κ
σ	deterministic selector for one reported optimum
θ	parameter along a named joint-law family
ξ	signal accuracy
ϕ	operational-flexibility or buffering parameter

2 Full finite-scenario linear programme

For scenario probabilities $w_m > 0$ summing to one, the finite-scenario formulation is

$$\max_{x,v,q} \quad \mu^\top x \tag{1}$$

$$\text{s.t.} \quad \mathbf{1}^\top x \leq A, \tag{2}$$

$$c^\top x \leq B, \tag{3}$$

$$Rx \leq r, \quad -Kx \leq -k, \quad Gx \leq h, \tag{4}$$

$$\ell \leq x \leq u, \tag{5}$$

$$v + \frac{1}{1-\alpha} \sum_{m=1}^M w_m q_m \leq \kappa, \tag{6}$$

$$-\pi_m^\top x - v - q_m \leq 0, \quad m = 1, \dots, M, \tag{7}$$

$$q_m \geq 0, \quad v \in \mathbb{R}. \tag{8}$$

The objective contains expected profit only; risk and operational effects enter through the explicit constraints.

2.1 Heuristic-policy comparison set and projection

For raw heuristic x^h , the principal comparison allocation is

$$\hat{x}_2^h = \arg \min_x \frac{1}{2} \|x - x^h\|_2^2$$

subject to $\mathbf{1}^\top x = 1$, the crop lower and upper bounds, operating budget, corn rotation cap, soybean contract minimum, planting-labour capacity and harvest-equipment capacity. Idle land is not allowed. The loss-CVaR ceiling is not included in this projection set; projected policies are evaluated against it afterwards. The unweighted objective is strictly convex, so the Euclidean projection is unique. SLSQP uses objective tolerance 10^{-10} , followed by an independent 10^{-8} feasibility check.

The sensitivity projection minimizes $\|x - x^h\|_1$ over the same set by HiGHS. A second linear programme selects one point on a non-unique L^1 -optimal face by minimizing crop-order weights $(1, 2, 3)^\top x$ within 10^{-10} of the minimum distance. This deterministic rule affects only benchmark comparison policies, not any optimized model solution.

2.2 KKT equations

Let η be the loss-CVaR-ceiling multiplier, t_m the multiplier for scenario excess row m , and z_m the multiplier on $q_m \geq 0$. Together with the operational multipliers defined in the main text, stationarity is

$$0 = -\mu + \nu^A \mathbf{1} + \nu^B c + R^\top \nu^R - K^\top \nu^K + G^\top \nu^G - \nu^\ell + \nu^u - \sum_m t_m \pi_m, \quad (9)$$

$$0 = \eta - \sum_m t_m, \quad (10)$$

$$0 = \frac{\eta w_m}{1 - \alpha} - t_m - z_m, \quad m = 1, \dots, M. \quad (11)$$

All multipliers associated with inequality constraints are nonnegative. Because v is unrestricted in sign, stationarity with respect to v yields the equality in Eq. (10). Complementarity requires each multiplier times its primal slack to be zero. The scenario equations allow fractional mass at the VaR threshold and are valid for discrete scenario distributions.

3 Proofs

3.1 Proof of the restricted reversal theorem

Let $z = x_1$ and $x_2 = A - z$. Expected profit is

$$\mu_1 z + \mu_2 (A - z) = \mu_2 A + (\mu_1 - \mu_2) z.$$

Because $\mu_1 > \mu_2$, this function is strictly increasing in z . Loss-CVaR is convex and continuous in the portfolio weights. Its sublevel set intersected with the compact operational interval I is therefore a compact interval $\mathcal{K}_{\theta, \kappa}$. A strictly increasing linear objective has the unique maximizer $\sup \mathcal{K}_{\theta, \kappa}$.

The pair reverses when $z < A - z$, equivalently $z < A/2$. Hence reversal holds if and only if the right endpoint of the entire risk-feasible interval lies below $A/2$, which is equivalent to no feasible point on $[A/2, \bar{z}]$. If risk is nondecreasing on that upper half, the minimum upper-half risk occurs at $A/2$, so the upper half is infeasible if and only if $r_\theta(A/2) > \kappa$. Existence of a feasible lower-half point ensures that this inequality indicates reversal rather than infeasibility.

If $\theta \mapsto r_\theta(A/2)$ is continuous, strictly increasing and crosses κ , the intermediate-value theorem gives a unique θ^* . If strictness is removed, an interval of thresholds can result. If monotonicity is removed, the reversal set can be disconnected. If crossing is removed, a threshold need not exist.

3.2 Proof of complete optimality

The finite-scenario problem is a bounded linear programme whenever $\mathcal{X}$ is nonempty and bounded. Linear-programming strong duality implies that primal and dual feasibility plus equality of the objective values are necessary and sufficient. Expanding equality of values across all inequality rows yields stationarity and complementarity. The equations above therefore form a complete certificate.

For the population convex formulation, write the Lagrangian using loss-CVaR as a convex function. Under Slater’s condition or another convex constraint qualification, KKT conditions are necessary; convexity makes them sufficient. Pairwise subtraction of acreage stationarity cancels ν^A but leaves budget, rotation, contract, shared-resource and bound terms. No algebraic operation on this local equality recovers an acreage ordering without the global feasible geometry.

3.3 Proof of named-family ordering

For integrable random losses, loss-CVaR is a law-invariant coherent risk measure and is monotone under convex order. Thus $L_{\theta_1}(x) \preceq_{\text{cx}} L_{\theta_2}(x)$ implies the stated loss-CVaR inequality at each x . Pointwise inequality gives containment of the two risk sublevel sets. Maximizing the same expected-profit objective over nested sets gives the optimal-value order. Coordinate monotonicity does not follow because the supporting face can change when the feasible set contracts.

3.4 Proofs for information and flexibility

In a posterior-separable model, or when the shared ex-ante risk restriction is nonbinding, any no-information action defines a constant signal-contingent policy. If the constant policy is feasible, optimization after observing a signal cannot lower value. When one common action is posterior-optimal for every signal, the constant policy also attains the informed optimum, proving zero value. If unique posterior optima differ on events with positive probability and at least one difference is strict, expected improvement is positive. This posterior-optimum strictness step is not applied when contingent actions share a binding ex-ante loss-CVaR budget.

Increasing differences and nested action sets yield monotone posterior actions under standard monotone comparative statics. Strictness on a positive- probability event makes the information gain larger at the higher acreage- flexibility level. This is the conditional complementarity result.

For buffering, at $\phi = 1$ all state margins equal $\bar{\pi}$; posterior optimization problems are identical and the common-action condition gives zero information value. If information value is positive at $\phi = 0$ and continuous along the contraction path, it must decline somewhere. Those intervals are substitutive. Total optimized value may still rise with ϕ .

For the complete shared ex-ante loss-CVaR numerical programme, interaction is classified without extending that theorem:

$$\Delta_{\xi,\phi}V = [V(\xi_2, \phi_2) - V(\xi_1, \phi_2)] - [V(\xi_2, \phi_1) - V(\xi_1, \phi_1)].$$

The sign is computed on each adjacent grid rectangle.

4 Calibration and constraint table

Total land is 1.0, operating budget is US\$330, planting-labour capacity is 0.82 and harvest-equipment capacity is 0.88. The full model uses $\alpha = 0.95$, $\rho = 0.5$, 512 optimization scenarios and seed 202607270034.

5 Policy comparison

The expected-profit endpoint is reported as an endpoint rather than as a mean–variance comparison policy. The diversification comparison uses one selected Gaussian mean–variance policy. Its x^0 is the

Table 2: Kansas calibration and primary allocation.

Crop	performance index	mean margin real 2024 US\$ per acre	margin s.d.	full-model share
Corn	0.748	199.8	94.9	0.330
Soybean	0.756	229.0	104.0	0.422
Winter wheat	0.894	138.3	49.6	0.248

Table 3: Operational set. Coefficients use normalized land and capacity units.

Object	Corn	Soybean	Winter wheat
Lower bound	0.05	0.10	0.05
Upper bound	0.55	0.65	0.60
Rotation cap	0.50	—	—
Contract minimum	—	0.10	—
Planting-labour coefficient	1.00	0.72	0.58
Harvest-equipment coefficient	1.05	0.82	0.62

matched-Gaussian expected-profit allocation; all 301 points over $\gamma \in [0, 0.03]$ are solved; and x^{MV} is the first point to attain the fixed 15% Gaussian variance-reduction target. At that point the reduction is 15.3%, the allocation differs from x^T , and Student- t evaluation loss-CVaR is \$2.364 higher. Both the weak and operationally strong diversification-failure criteria hold for $\gamma \in [0.0068, 0.0088]$. The two classifications share the same interval in this experiment because the tail-aware optimum binds the common loss-CVaR ceiling.

The alternative L^1 projection leaves the already feasible score-proportional and equal-share policies unchanged. It maps winner take all to (0.300, 0.100, 0.600) at distance 0.800, compared with the Euclidean allocation (0.200, 0.200, 0.600). Both are risk-feasible, but only the L^1 allocation reverses the soybean–corn pair. The score-winner reversal classification is therefore projection-sensitive.

One-factor sensitivity contains 18 cases covering scenario count, seed, Kendall’s τ , Student- t copula degrees of freedom, loss-CVaR confidence level, risk-ceiling location, evaluation-law marginal specification and the 10%, 15% and 20% selection targets. 17 cases satisfy the selected-policy weak and strong criteria. Under the 20% target, the selected value $\gamma = 0.0089$ lies just beyond the focal strong-failure interval; its positive tail gap is smaller than the declared risk tolerance ε_τ .

6 Phase-diagram summary

The mechanism-isolation analysis covers every evaluated parameter combination. In the margin layer, top-ranked winter wheat has the lowest public-data mean margin and the expected-profit endpoint already has a complete rank reversal; loss-CVaR raises its share. In the risk layer, soybean has the higher index and mean margin, while a mean-preserving 10% downside event of \$154.7 per acre produces the loose-to-tight allocation crossing reported in the main text. Across the probability-by-magnitude map, 24 of the 49 stress combinations produce an allocation crossing, 24 remain feasible without a crossing, and one is infeasible. In the operational layer, the baseline calibrated constraint sequence remains non-reversing before the soybean rotation-cap path is evaluated; the first crossing is at 0.35. These controlled paths demonstrate mechanisms rather than their population frequency.

Across all 165 principal cells, selected pairwise reversal occurs in 143 and selected complete rank reversal in 96. Strong exclusion is structurally inadmissible because every crop has a positive lower bound. Complete family-by- τ tables report all three ranked-pair face extrema and possible, universal and selected classifications for every parameter combination.

No selected, possible or universal exclusion is found after crop exit becomes admissible. Hence no first

Table 4: Primary policy outcomes. A larger loss-CVaR is worse.

Policy	Corn	Soy	Wheat	Profit	loss-CVaR	risk feasible
Score proportional, Euclidean	0.312	0.315	0.373	187.4	-53.1	yes
Winner take all, Euclidean	0.200	0.200	0.600	169.6	-59.4	yes
Equal share	0.333	0.333	0.333	190.4	-51.4	yes
Expected profit	0.280	0.650	0.070	216.5	-34.5	no
Full loss-CVaR	0.330	0.422	0.248	198.5	-47.0	yes
Minimum loss-CVaR endpoint	0.220	0.180	0.600	169.0	-59.5	yes

Table 5: Selected Gaussian mean–variance policy. Operational feasibility applies to the common full-investment set; risk feasibility is evaluated under the Student- t law.

γ	Corn	Soybean	Winter wheat	operational/risk feasible
0.0082	0.314	0.543	0.143	yes/no
Gaussian mean profit		Gaussian variance		Student- t loss-CVaR
206.6		5733.5		-43.4

strong exclusion reversal boundary, excluded crop or active constraint at an exclusion transition exists on the evaluated grid. The minimum selected highest-ranked-crop share under all-zero bounds is 0.070.

7 Robustness and uncertainty

The one-at-a-time robustness analysis reports selected pairwise and selected complete rank reversal classifications separately for confidence level, constraint regime, dependence family and parameter, marginal law, sample window, scenario count, random seed and solver. Its strong classification is structurally unavailable because all crop lower bounds are positive. In the all-zero and highest-ranked-crop-zero phases, strong exclusion reversal remains absent for every zero tolerance from 10^{-8} to 10^{-2} ; pairwise and complete rank reversal counts change only at the coarser ordering tolerances.

The 64 historical bootstrap replications produce mean shares 0.277 for corn, 0.527 for soybean and 0.196 for winter wheat. Their 95% percentile intervals are [0.125, 0.368], [0.258, 0.650] and [0.070, 0.404]. The binary reversal frequency across historical resamples is treated separately. Selected pairwise reversal occurs in 62 of 64 resamples. Its exact Clopper–Pearson binomial interval is [0.892, 0.996]. The interval describes the frequency among historical resamples, not a farm-population probability. The first-reversal index mean is 0.066 with interval [0.000, 0.598]. The broad boundary interval is consistent with an eight-year calibration; simulated scenarios do not add independent historical information.

8 Information–flexibility details

The state split contains four low and four high relative-yield-advantage years. Signal accuracy takes six values and flexibility takes six values on each path, producing 72 cells. Adjacent-grid classifications are 32 positive cross-differences, 17 negative cross-differences, 18 zero-information and 5 zero/boundary. Every signal-contingent problem is feasible and the minimum computed value of information differs from zero only at floating-point scale.

At perfect signal accuracy, post-signal acreage reallocation raises information value monotonically from zero at no adjustable acreage to \$4.655 at full adjustment. Under shock buffering, the corresponding information value declines toward zero as the state margins are contracted to a common vector. This comparison is about interaction; buffering can still increase the no-information optimized value.

Table 6: Raw heuristics and principal Euclidean projections. Distances are unweighted Euclidean distances in normalized land share.

Policy	Raw			Projected			distance L^2
	Corn	Soy	Wheat	Corn	Soy	Wheat	
Score proportional	0.312	0.315	0.373	0.312	0.315	0.373	0.000
Winner take all	0.000	0.000	1.000	0.200	0.200	0.600	0.490
Equal share	0.333	0.333	0.333	0.333	0.333	0.333	0.000

Table 7: Lower-bound sensitivity at numerical-zero tolerance 10^{-4} . The principal strong exclusion count is structurally unavailable; the relaxed specifications permit exclusion.

Lower-bound specification	pairwise reversal	complete rank reversal	selected strong exclusion	possible strong exclusion	universal strong exclusion	multiple/ infeasible
Positive crop minima	143	96	structural	structural	structural	2/0
All crop minima zero	143	95	0	0	0	2/0
Highest-ranked crop minimum zero	143	96	0	0	0	2/0

9 External panel and identification boundary

Table 8: Top-rank disagreement in 248 state-years.

Ranking definition	estimate	cluster 95% low	cluster 95% high
Relative yield	0.887	0.802	0.954
Standardized revenue	0.613	0.447	0.767
Operating margin	0.625	0.552	0.693
Total-cost margin	0.782	0.701	0.857

The leakage-free relative-yield specification uses 651 crop-transition rows. Its prior-leader effect is -0.0042 with interval [-0.0123,0.0056]. The estimand is descriptive because the panel does not observe farm objectives, local prices or costs, credit, rotation history, contracts, risk preferences or received forecasts. The state acreage panel is therefore used for descriptive comparison rather than as an optimized allocation.

10 Data availability

The analysis uses public USDA NASS, USDA ERS and US Bureau of Labor Statistics files joined by crop and year. The public source reports, analysis-ready tables and figure source data needed to reproduce the reported results will be deposited in a public archive upon publication. No confidential farm-level data are used.

11 Code availability

Analysis code sufficient to regenerate the numerical results and figures will be deposited with the source data upon publication.

Relaxing crop lower bounds leaves strong exclusion at zero

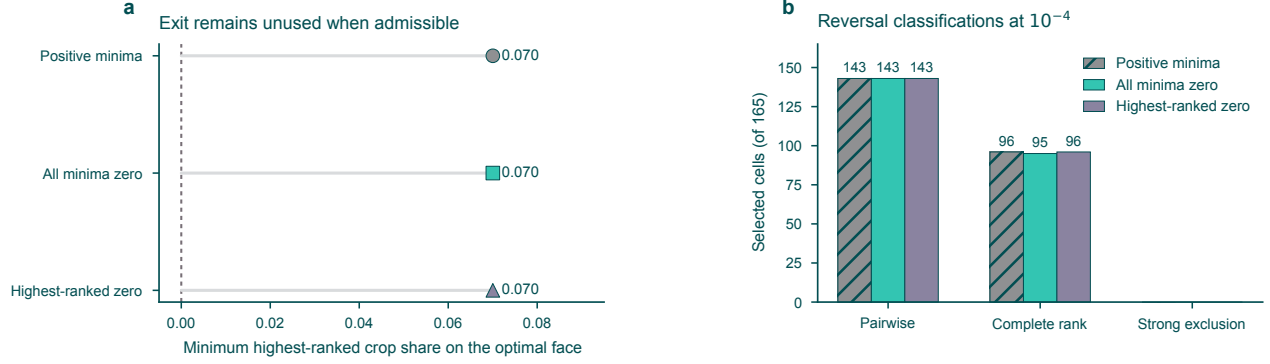


Figure 1: Lower-bound sensitivity makes crop exit admissible but does not produce strong exclusion reversal on the evaluated grid. **a**, Minimum winter-wheat allocation over the complete optimal face under the principal, all-zero and highest-ranked-crop-zero lower-bound specifications. All remain bounded away from numerical zero. **b**, Selected pairwise reversal, complete rank reversal and strong exclusion reversal counts at tolerance 10^{-4} . Hatching marks the principal strong count as structurally unavailable; neither relaxed specification produces strong exclusion reversal. All 165 parameter combinations are shown for each specification.

12 Limitations

1. The Kansas calibration has eight effective annual observations. It cannot estimate extreme-tail dependence reliably.
2. State yield combined with national price and cost is not a farm-level profit series.
3. Copula parameters and shared-capacity limits are controlled stress values, not causal estimates.
4. The model is single-period and linear. It omits fixed charges, endogenous prices, spatial spillovers and explicit multi-year rotation state.
5. Historical bootstrap replications quantify sensitivity to the short calibration record but do not create independent empirical observations.
6. The information state is a transparent annual proxy, not a deployed forecast product.
7. The external panel is aggregate and descriptive; it cannot identify the full model's active mechanism or welfare.